

LAB EXPERIMENT RECORD

Department	AIDS
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R
Name	DEEKSHITHA U
Register Number	192424055
Date	08 October 2026
Day	Thursday

SET 14: Survey Results Analysis

Aim

To analyze survey response data using a **stacked bar chart, radar chart, table, and Tableau dashboard** for interactive exploration of survey results.

Dataset

Survey Results

Respondent	Question 1	Question 2	Question 3
1	A	B	C
2	B	A	D
3	C	A	B

Q1. Stacked Bar Chart

Question

Create a stacked bar chart to display the distribution of answers for Question 1. Label the chart elements.

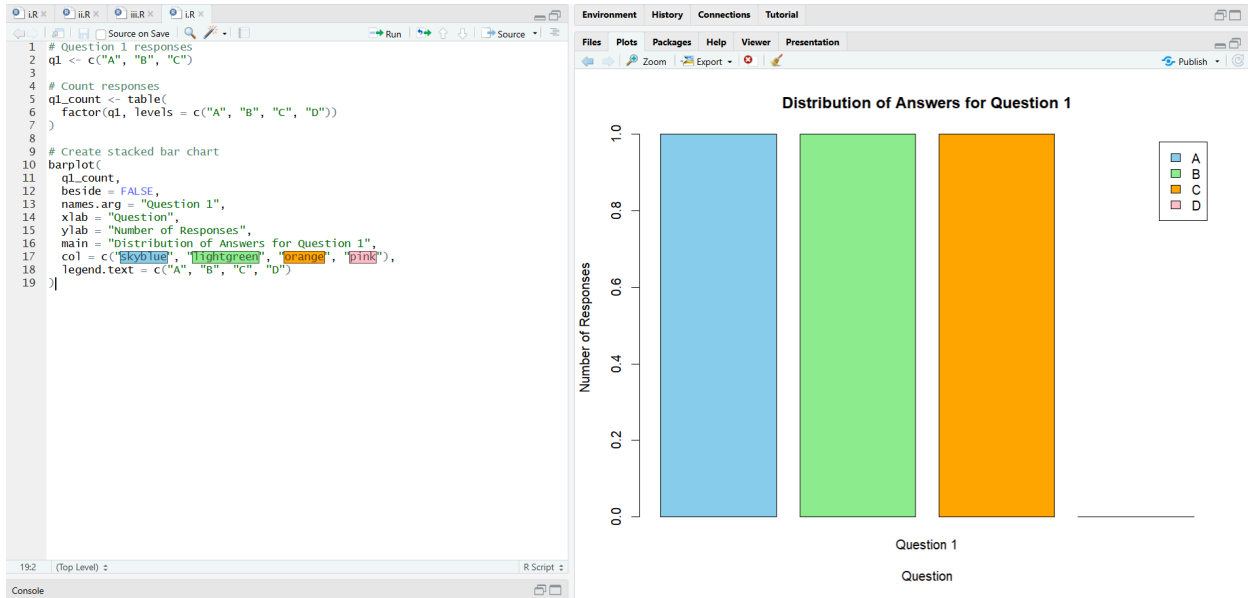
R Program

```
q1 <- c("A", "B", "C")

# Count responses
q1_count <- table(
  factor(q1, levels = c("A", "B", "C", "D"))
)

# Create stacked bar chart
barplot(
  q1_count,
  beside = FALSE,
  names.arg = "Question 1",
  xlab = "Question",
  ylab = "Number of Responses",
  main = "Distribution of Answers for Question 1",
  col = c("skyblue", "lightgreen", "orange", "pink"),
  legend.text = c("A", "B", "C", "D")
)
```

Output



Observation

Question 1 has an equal distribution of **A, B, and C**, with each answer selected once.

Q2. Radar Chart — Overall Response Pattern

Question

Generate a radar chart to visualize the overall pattern of responses across all three questions.

Since the responses are categorical, we first calculate the **number of A, B, C, and D responses for each question**.

R Program

```
# Install package once if needed:
# install.packages("fmsb")

library(fmsb)

# Response data
q1 <- c("A", "B", "C")
q2 <- c("B", "A", "A")
q3 <- c("C", "D", "B")

# Count each response for every question
response_data <- rbind(
  Q1 = table(factor(q1, levels = c("A", "B", "C", "D"))),
  Q2 = table(factor(q2, levels = c("A", "B", "C", "D"))),
  Q3 = table(factor(q3, levels = c("A", "B", "C", "D")))
)

# Maximum and minimum values required by fmsb
radar_data <- rbind(
  Max = c(3, 3, 3, 3),
  Min = c(0, 0, 0, 0),
  response_data
)

# Radar chart
radarchart(
  radar_data,
  axistype = 1,
  pcol = c("blue", "red", "green"),
  plwd = 2,
```

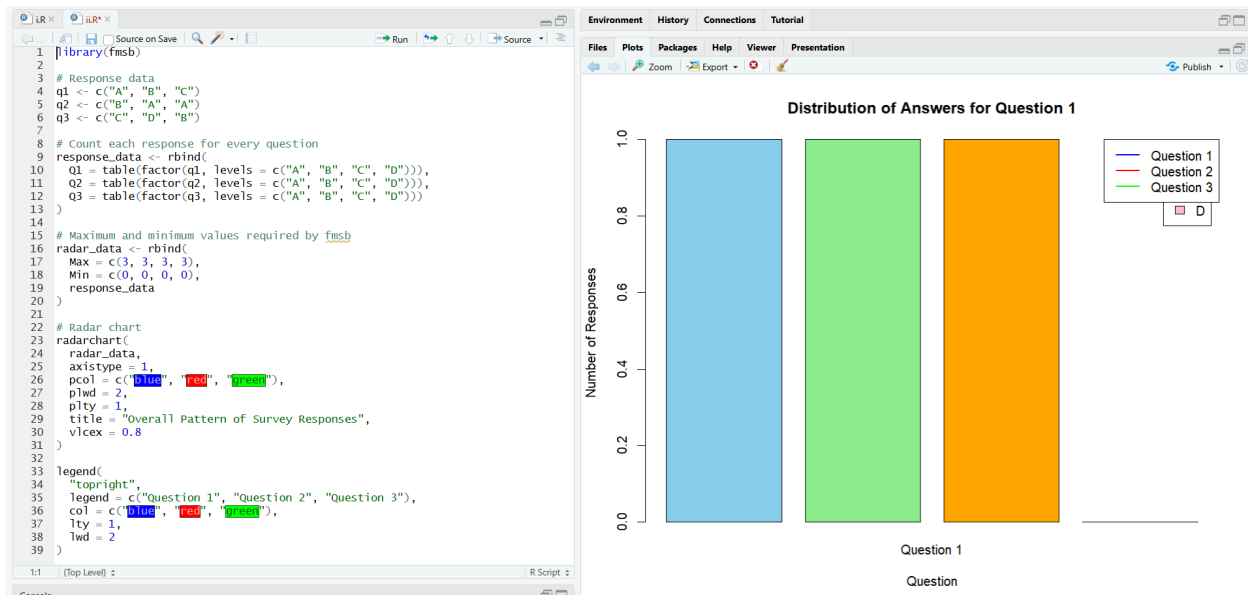
```

    pty = 1,
    title = "Overall Pattern of Survey Responses",
    vlcex = 0.8
  )

  legend(
    "topright",
    legend = c("Question 1", "Question 2", "Question 3"),
    col = c("blue", "red", "green"),
    lty = 1,
    lwd = 2
  )

```

Output



Observation

- Question 1 contains A, B, and C once each.
- Question 2 contains A twice and B once.
- Question 3 contains B, C, and D once each.
- The radar chart makes these differences in response patterns visually comparable.

Q3. Survey Response Table

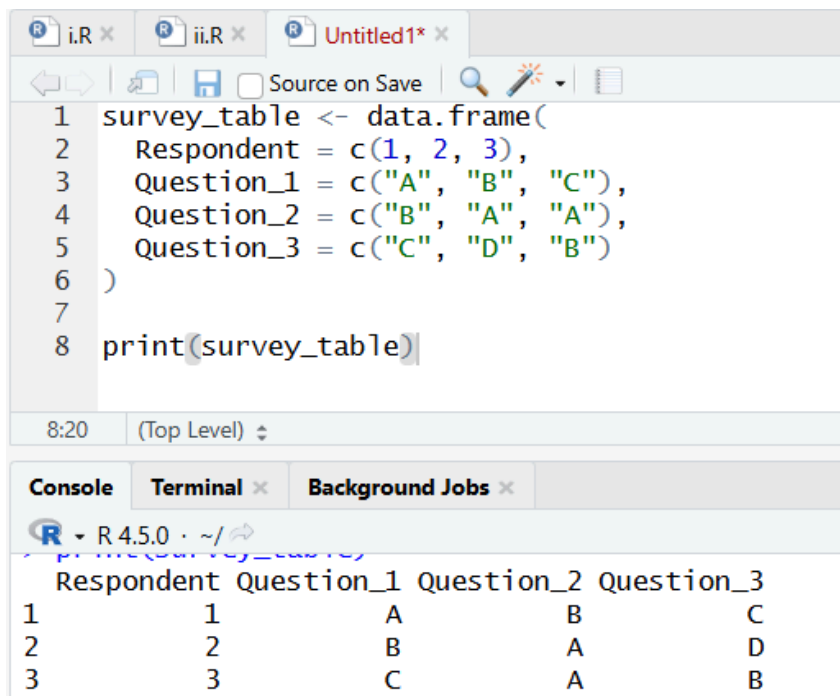
Question

Build a table to show the survey response data for each respondent. Label the table elements.

R Program

```
survey_table <- data.frame(  
  Respondent = c(1, 2, 3),  
  Question_1 = c("A", "B", "C"),  
  Question_2 = c("B", "A", "A"),  
  Question_3 = c("C", "D", "B")  
)  
  
print(survey_table)
```

Output



The screenshot shows the R Studio environment. The script editor contains the following code:

```
1 survey_table <- data.frame(  
2   Respondent = c(1, 2, 3),  
3   Question_1 = c("A", "B", "C"),  
4   Question_2 = c("B", "A", "A"),  
5   Question_3 = c("C", "D", "B")  
6 )  
7  
8 print(survey_table)
```

The console output shows the resulting data frame:

```
Respondent Question_1 Question_2 Question_3  
1          1          A          B          C  
2          2          B          A          D  
3          3          C          A          B
```

Observation

The table successfully displays the responses given by each respondent for all three questions.

Q4. Tableau Dashboard

Question

Develop a Tableau dashboard combining the stacked bar chart, radar chart, and table for interactive exploration of survey responses.

Step 1 — Prepare the Data

Create an Excel/CSV file:

Respondent	Question 1	Question 2	Question 3
1	A	B	C
2	B	A	D
3	C	A	B

Step 2 — Import Dataset

1. Open **Tableau**.
2. Select **Microsoft Excel** or **Text File**.
3. Import the survey results dataset.

Step 3 — Create Stacked Bar Chart

- Create a worksheet for Question 1.
- Drag **Question 1** → Columns.
- Drag **Respondent** → Rows.
- Use **Count** of Respondent.
- Drag **Question 1** → Color.
- Enable **Stack Marks**.
- Add labels.

Title: Question 1 Response Distribution

Step 4 — Create Radar Chart

Tableau does not provide a standard radar chart directly in its basic chart types.

For a radar chart, prepare a summarized response-frequency table containing:

Question	A	B	C	D
Question 1	1	1	1	0
Question 2	2	1	0	0
Question 3	0	1	1	1

Then use a radar-chart construction or Tableau-compatible radar-chart technique.

Title: Overall Pattern of Survey Responses

Step 5 — Create Response Table

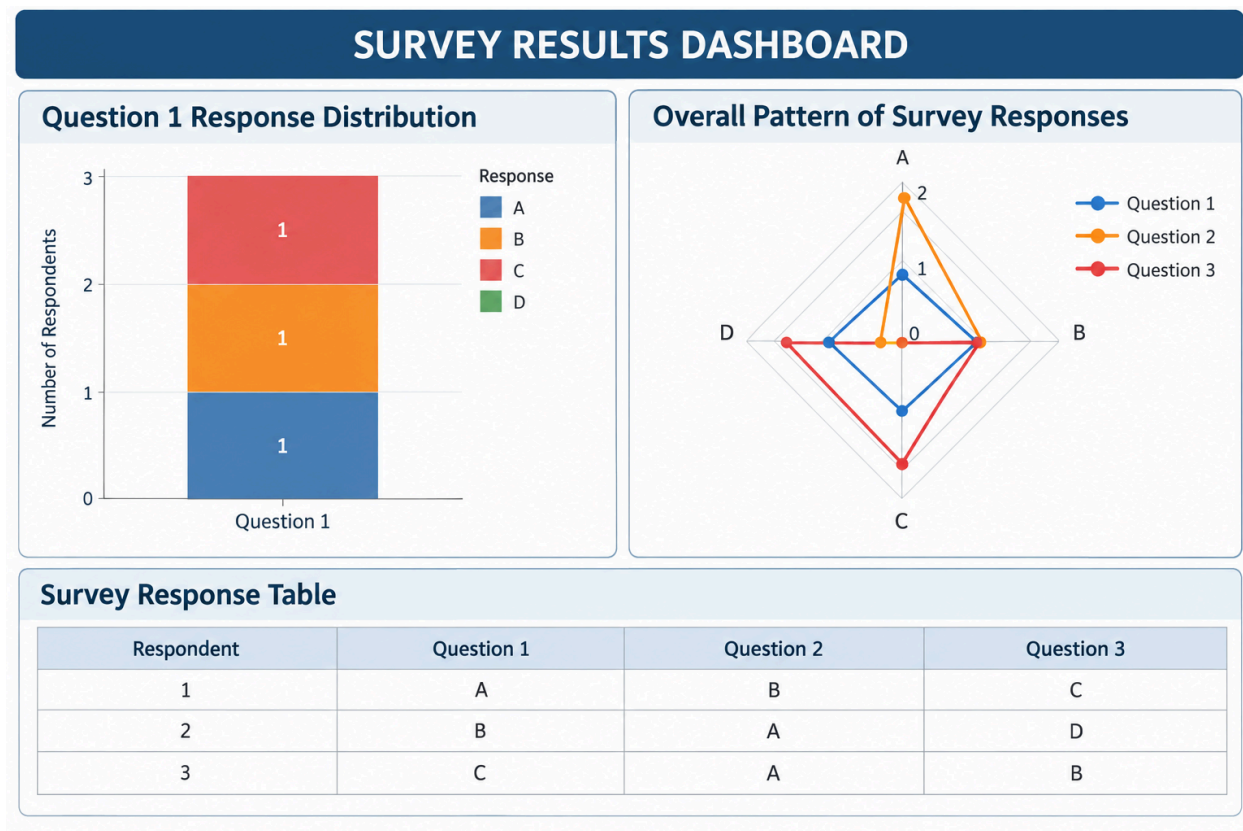
Create a worksheet containing:

- Respondent
- Question 1
- Question 2
- Question 3

Step 6 — Create Dashboard

1. Select **New Dashboard**.
2. Add the **Stacked Bar Chart**.
3. Add the **Radar Chart**.
4. Add the **Survey Response Table**.
5. Add filters if required.
6. Arrange the visualizations neatly.

Expected Dashboard



Procedure

1. Open **RStudio** and enter the given survey response data.
2. Extract the responses for Question 1.
3. Use the `table()` function to calculate the frequency of each answer.
4. Use `barplot()` with `beside = FALSE` to create the **stacked bar chart**.
5. Calculate the response frequencies for A, B, C, and D across all three questions.
6. Use the `fmsb` package and `radarchart()` function to create the **radar chart**.
7. Use `data.frame()` to create the survey response table.
8. Open **Tableau** and import the survey results dataset.
9. Create the stacked bar chart for Question 1.
10. Create the radar chart using the summarized response-frequency data.
11. Create a table showing the responses for each respondent.
12. Combine the three visualizations into a **Tableau dashboard**.
13. Add suitable titles, labels, filters, and formatting for interactive exploration.

Result

The survey results were successfully analyzed using **R and Tableau**. A stacked bar chart was created to display the distribution of Question 1 responses, a radar chart was created to visualize the overall response pattern across the three questions, and a table was created to display individual respondent responses. These visualizations were successfully combined into an **interactive Tableau dashboard**.